

AN ENTERPRISE ARCHITECTURE FOR THE AGENTIC ERA

AIDEX

*An Architecture for Human-Curated,
AI-Enabled Knowledge Work*

James D. Longmire

Northrop Grumman Fellow · Chief Architect, Digital Ecosystems

Independent research · ORCID 0009-0009-1383-7698

Three transitions are happening at once.

01

Workforce

Senior engineers retiring faster than they can be replaced. Mid-career talent recruited across boundaries. Early-career hires entering programs whose context took fifteen years to absorb.

02

Substrate

The tools available to the individual knowledge worker in 2026 differ in kind, not in degree. Agentic execution, queryable synthesis, orchestrated workflow. The shape of work has changed.

03

Accountability

Consumer-grade AI does not solve accountability. The defense enterprise that builds substrate without enforced human accountability has deployed a system the institution cannot govern.

The enterprise that deploys deliberately gets the strategic position.

Four terms that work together.

AEON

AI Enterprise Orchestration Nexus

Enterprise control plane. Identity, authority, policy, evidence, integration, capability composition.

AIDEX

AI Digital Experience

Worker-facing subdomain within AEON. The substrate on which the knowledge worker operates.

HCAE

Human-Curated, AI-Enabled practice

The discipline AIDEX expresses. Practitioner curates AI-enabled output, remains the locus of judgment and accountability.

The Practitioner

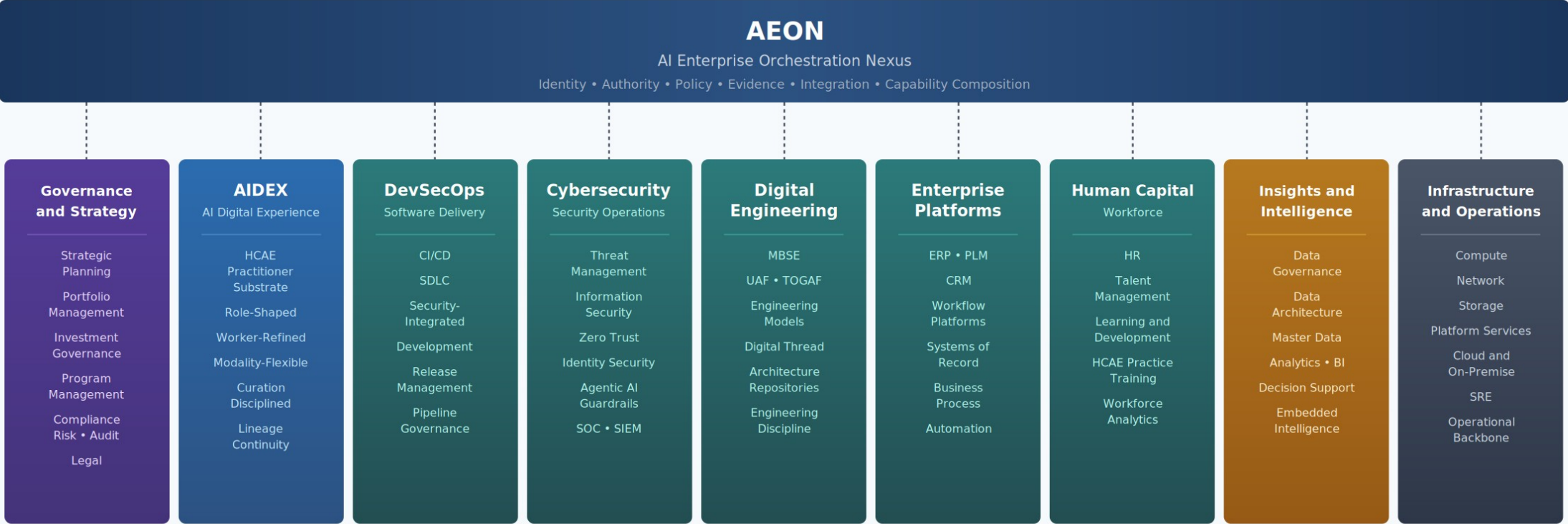
Next-generation knowledge worker

The human protagonist. HCAE practitioner operating on AIDEX under AEON orchestration.

Nine peer subdomains under one control plane.

AEON Enterprise Architecture Viewpoint

AI Enterprise Orchestration Nexus over Peer Subdomain Capabilities



Subdomains operate as peers under AEON orchestration. Identity, policy, evidence, and integration fabric flow across all subdomains.
HCAE is the practice framework AIDEX expresses; HCAE-aligned discipline extends to worker engagement with all subdomains.

AIDEX is one peer subdomain among nine. AEON orchestrates capability composition across all of them.

HCAE is the discipline. The order matters.

Human-curated first. Curation is the act that anchors accountability.

AI-enabled second. Enablement is the function of the substrate.

The practitioner is the principal. The substrate is the instrument. Reversing the order produces a different framework.

Risk-tiered curation

Curation discipline applied at the right level for the consequence of the action.

PER-ACTION

High-consequence actions. Explicit confirmation captured on each.

BATCH

Medium-consequence. Worker confirms grouped actions at end of unit of work.

SAMPLED AUDIT

Low-consequence routine. Audit sampling at the evidence plane.

What AIDEX does for the practitioner.

Role-shaped

The substrate is configured for the specific role the practitioner occupies. Architect, engineer, program manager, analyst. Different roles see different capabilities, different defaults, different policies.

Worker-refined

The practitioner shapes her own substrate over time. Memory layer is worker-owned. Personalization compounds with tenure.

Modality-flexible

Text, voice, ambient, programmatic. The substrate participates across surfaces while preserving identity, authority, and lineage continuity.

Curation-disciplined

Propose-and-confirm posture at every consequential action. Captured curation becomes part of the evidence chain AEON aggregates.

Eight axes (presentation, persona, role, authority, context, memory, modality, lineage) vary independently. Detail follows.

Eight axes vary independently.

01

Presentation

Named entity, voice, visual presence.
Worker-selectable where appropriate.

02

Persona

Communication register and pattern.
Configured by deployment.

03

Role

Functional taxonomy that shapes
available capabilities.

04

Authority

AEON-evaluated policy on identity-
resolved attributes.

05

Context

Program, project, and situational scope
of the work.

06

Memory

Worker-owned personalization. Opt-in,
attributed, revocable.

07

Modality

Text, voice, ambient, programmatic.
Identity continuous across.

08

Lineage

Attributed contribution to institutional
practice over tenure.

AEON provides. AIDEX consumes.

AIDEX does not maintain its own user store, policy engine, audit infrastructure, or vendor governance. AEON provides these as enterprise services.

Identity

Authenticated principal across classification environments. Federation that AIDEX inherits.

Authority

Policy evaluation on identity-resolved attributes at moment of action.

Policy

Decomposition from enterprise scope into subdomain-specific enforcement.

Evidence

Aggregation of subdomain activity into enterprise audit picture.

Integration

Cross-subdomain workflow orchestration. AIDEX agents participate.

Composition

Capability lifecycle. What can be combined, by whom, for what purpose.

Claude Cowork shows the productivity layer is real.

Released January 2026 (research preview), GA April 2026. The interaction substrate is configurable, not developable. The remaining work concentrates in AEON integration, role packaging, classification topology, and lineage.

WHAT COWORK DEMONSTRATES

- Agentic interaction at scale, deployable today
- Plugin architecture for role-specific capability
- Enterprise governance posture sufficient for many contexts
- Multi-modality across desktop and ambient surfaces
- Memory layer that personalizes over time

WHERE IT FALLS SHORT ON HCAE

- No deep AEON IDAM binding or delegation chain enforcement
- Memory not yet a long-arc role-shaped accumulation
- No lineage layer for attributed institutional contribution
- Partial classification deployment story
- Vendor-coupled; substrate portability requires architectural work

Multi-backend by design.

The conventional cloud-versus-on-prem framing is wrong. The real picture has four topologies mapped to different classification environments. AIDEX supports all four simultaneously.

Commercial Cloud

UNCLASSIFIED

Frontier-class capability. Vendor-hosted. Suitable for non-sensitive enterprise work.

GovCloud

CUI / FedRAMP HIGH

Frontier capability within accredited boundaries. The mainstream defense deployment target.

On-Premises Open-Weight

NOFORN / RESTRICTED

Llama-class models inside the boundary. Capability gap acknowledged. Air-gap compatible.

On-Premises Commercial

EMERGING

Vendor models behind enterprise control. Currently rare; trajectory worth tracking.

The accountability frontier.

Consumer-grade AI has not had to solve accountability. The consumer market does not require it. The defense enterprise that deploys substrate without enforced human accountability has deployed a system the institution cannot govern.

01

Consumer-grade alternatives are not building here.

02

The defense enterprise needs this regardless of AI vendor strategy.

03

Early HCAE-aligned deployment compounds the institutional advantage.

This is not just an IT program.

Enterprise IT leads. Success depends on partnership across the institution.

HR	Worker-ownership and consent commitments. Employment, IP, and separation processes that operationalize HCAE practitioner rights.
Chief Learning Officer	HCAE practice training. Curation discipline development. Workforce capability building for the agentic substrate.
Program Leadership	Deployment selection and operational integration. The substrate has to fit how programs actually run.
Information Security	AEON control plane integration. Identity federation across classification environments. Cryptographic delegation chains.
Legal	Contractual and policy infrastructure for worker contributions, revocation rights, records retention, and accountability allocation.

What we are asking.

Sanction a pilot.

TARGET ROLE	One well-defined role with clear curation expectations.
TARGET PROGRAM	One program with engaged leadership and tractable systems of record.
TARGET TOPOLOGY	Whichever classification environment matches the program.
PARTNERSHIPS	HR, CLO, Information Security, Legal aligned before deployment.
INSTRUMENTATION	Metrics defined that can falsify the strategic argument, not just confirm it.

The workforce transition is happening on its own timeline. The question is whether the enterprise faces it with the framework, the subdomain, and the control plane, or without them.